



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023-2024)

Test Pattern

TG: @Chalnaayaaar NEET (UG)

MINOR

16-07-2023

PRE-MEDICAL : ENTHUSIAST ADVANCE COURSE PHASE - MEA & MEPS-I (A)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their **Form No.** in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

Test Booklet Code

This Booklet contains **28** pages.

E6

Do not open this Test Booklet until you are asked to do so.

Important Instructions :

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Biology-I and Biology-II)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below :
 - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 to 185). All questions are compulsory.
 - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150 and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject. **Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
- No candidate, without special permission of the Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of scribe or not.

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

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Your Target is to secure Good Rank in Pre-Medical 2024

Topic : MEC (CAPACITOR), WAVE THEORY (FROM INTERFERENCE), MODERN PHYSICS (TILL LAST LECTURE), [BACK UNIT] : CAPACITOR, UNIT DIMENSION AND ERROR

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SECTION-A (PHYSICS)

- Two waves of wave length 2 m and 2.02 m respectively moving with the same velocity and superimpose to produce 2 beats per second. The velocity of the waves is :
(1) 400.0 m/s (2) 402 m/s
(3) 404 m/s (4) 406 m/s
- The equation of stationary wave is $y = 10 \cos\left(\frac{\pi x}{10}\right) \sin(100\pi t)$ (where x is in cm and t is in s). The wave is formed on a string of length 40 cm. An antinode will be located at (from the following options)
(1) x = 15 cm (2) x = 25 cm
(3) x = 20 cm (4) x = 12.5 cm
- If a resonance tube gives two consecutive resonances at the length of 15 and 48 cm, then the velocity of sound in air is...
[frequency of fork = 500 Hz]
(1) 320 m/s (2) 330 m/s
(3) 340 m/s (4) 350 m/s
- Two vibrating strings of the same material but length L and 2L have radii 2r and r respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental modes, the one of length L with frequency n_1 and the other with frequency n_2 . The ratio $\frac{n_1}{n_2}$ is given by :-
(1) 2 (2) 4 (3) 8 (4) 1
- A soldier fires a bullet towards a hill hears the first echo after 4 s. He moves a distance 100 m towards the hill and fires a bullet and hears the echo after 3.5 s. The velocity of sound in air is
(1) 330 m/s (2) 300 m/s
(3) 400 m/s (4) 360 m/s
- An observer moves towards a stationary source of sound, with a velocity one – fifth of the velocity of sound. Which is the percentage increase in the apparent frequency?
(1) 5% (2) 20% (3) Zero (4) 0.5%
- A train moving at a speed of 220 ms^{-1} towards a stationary object, emits a sound of frequency 1000Hz. Some of the sound reaching the object gets reflected back to the train as echo. The frequency of the echo as detected by the driver of the train is :
(speed of sound in air is 330 ms^{-1})
(1) 5000 Hz (2) 3000 Hz
(3) 3500 Hz (4) 4000 Hz
- A rocket is going away from the earth at a speed of 10^6 m/s . If the wavelength of the light wave emitted by a light source on it is $5400 \text{ \AA}$, what will be its Doppler Shift?
(1) $1.8 \text{ \AA}$ (2) $18 \text{ \AA}$ (3) $25 \text{ \AA}$ (4) $14 \text{ \AA}$
- Monochromatic light of frequency $6 \times 10^{14} \text{ Hz}$ is produced by a laser. The power emitted is $2 \times 10^{-3} \text{ W}$. The no. of photons emitted per sec. will be :
(1) 5×10^{14} (2) 5×10^{15}
(3) 5×10^{16} (4) 5×10^{17}
- A beam of light of wavelength λ is totally reflected at normal incidence by a plane mirror. The intensity of the light is such that photons hit the mirror at a rate n. Given that the Planck constant is h, the force exerted on the mirror by this beam is :-
(1) $nh\lambda$ (2) nh/λ (3) $2nh/\lambda$ (4) $2n\lambda/h$
- If frequency of incident photons is doubled then stopping voltage will :-
(1) get double (2) be more than double
(3) less than double (4) remains same

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PHASE - MEA & MEPS

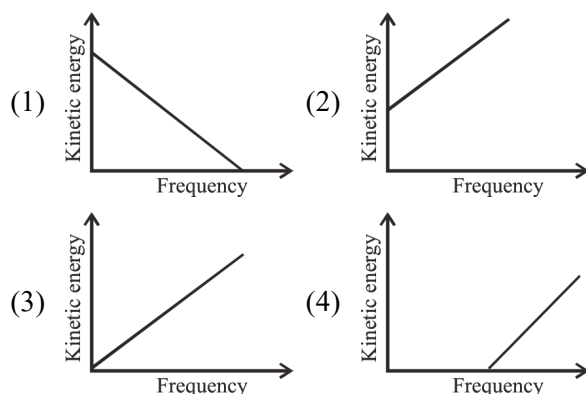
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12. A photosensitive metallic surface has work function ϕ . If photon of energy 3ϕ fall on this surface, the electron comes out with a maximum velocity of $6 \times 10^6 \text{ ms}^{-1}$. When the photon energy is increased to 9ϕ , there maximum velocity of photoelectron will be :-

- (1) $12 \times 10^6 \text{ m/s}$ (2) $6 \times 10^6 \text{ m/s}$
(3) $3 \times 10^6 \text{ m/s}$ (4) $24 \times 10^6 \text{ m/s}$

13. According to Einstein's photoelectric equation, the graph between the maximum kinetic energy of photoelectrons ejected and the frequency of incident radiation is



14. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are:-

- (1) 4, 4, 2 (2) 5, 1, 2
(3) 5, 1, 5 (4) 5, 5, 2

15. An object covers $(16.0 \pm 0.4)\text{m}$ distance in $(4.0 \pm 0.2)\text{s}$. Find out its speed :-

- (1) $(4.0 \pm 0.0) \text{ m/s}$ (2) $(4.0 \pm 0.2) \text{ m/s}$
(3) $(4.0 \pm 0.3) \text{ m/s}$ (4) $(20.0 \pm 0.6) \text{ m/s}$

16. A student measured the diameter of a rod by using screwgauge with pitch 1mm and 200 div on circular scale. In observation the main scale reading is 5 mm and 50 division above the reference level of circular scale then the correct diameter of ball is :-

- (1) 5.25 mm (2) 5.20 mm
(3) 5.200 mm (4) 5.250 mm

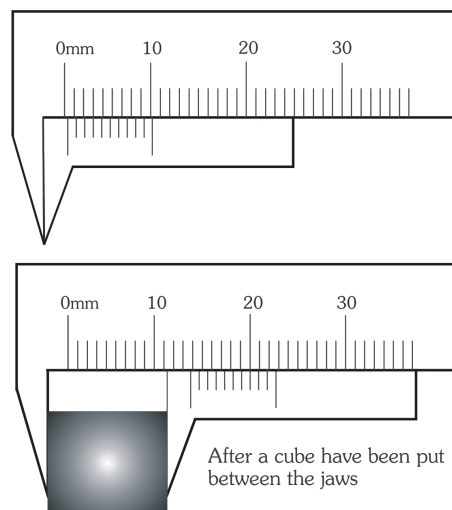
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PHASE - MEA & MEPS

17. If $P = \frac{\alpha}{\beta} e^{-\alpha t}$ where P is pressure and t is time α, β are constant find dimension of α :-

- (1) $[\text{ML}^{-1}]$ (2) $[\text{ML}^{-1}\text{T}^{-1}]$
(3) $[\text{M}^{-1}\text{LT}]$ (4) $[\text{T}^{-1}]$

18. Find the thickness of the cubical object using a defective vernier calliper. Main scale has mm marks and 10 divisions of vernier scale coincide with 9 divisions of main scale.



- (1) 13.8 mm (2) 13.4 mm
(3) 14.1 mm (4) 13.0 mm

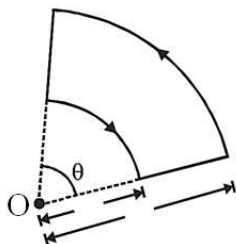
19. The plates of a parallel plate capacitor are completely filled by a solid dielectric. This capacitor is connected to a battery. Now the plates of the capacitor are pulled slightly apart. At the steady state,

- (1) the p.d. across the plates has increased.
(2) the energy stored by the capacitor has increased
(3) the capacitance of the capacitor has decreased.
(4) the charge on capacitor has increased

20. The current is flowing in south direction along a power line. The direction of magnetic field above the power line (neglecting earth's field) is:-

- (1) South (2) East
(3) North (4) West

21. A wire loop is formed by joining two sections of radii r_1 and r_2 subtending an angle θ at O. The magnetic field at O is B_0 :-

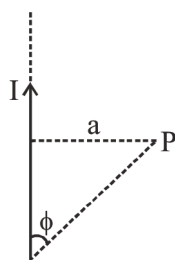


- (1) $B_0 = \frac{\mu_0 I}{4\pi} \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \theta$; outwards
 (2) $B_0 = \frac{\mu_0 I}{4\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \theta$; inwards
 (3) $B_0 = \frac{\mu_0 I}{2\pi} \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \theta$; outwards
 (4) $B_0 = \frac{\mu_0 I}{2\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \theta$; inwards

22. A straight section PQ of a circuit lies along the x-axis from $x = \frac{-a}{2}$ to $x = \frac{a}{2}$ and carries a steady current i . The magnetic field due to the section PQ at a point $x = +a$ will be :-

- (1) Proportional to a (2) Proportional to a^2
 (3) Proportional to $\frac{1}{a}$ (4) Zero

23. A semi-infinite straight conductor carries a current I . P is a point at perpendicular distance 'a' from the conductor as shown. The field at P due to the conductor is :



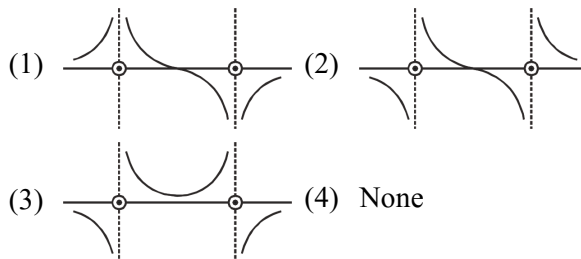
- (1) $B = \frac{\mu_0 I}{4\pi a} (1 + \sin \phi)$; outwards
 (2) $B = \frac{\mu_0 I}{4\pi a} (1 + \cos \phi)$; outwards
 (3) $B = \frac{\mu_0 I}{4\pi a} (1 + \sin \phi)$; inwards
 (4) $B = \frac{\mu_0 I}{4\pi a} (1 + \cos \phi)$; inwards

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PHASE - MEA & MEPS

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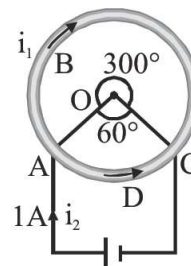
24. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper as shown in figure. The variation of magnetic field B along the line XX' is given by :-



25. A circular current carrying coil has a radius R . The distance from the centre of the coil on the axis where the magnetic induction will be $\frac{1}{8}$ th to its value at the centre of the coil, is :-

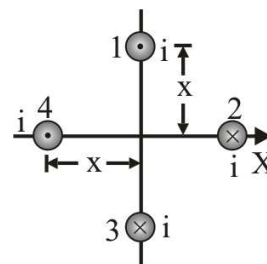
- (1) $\frac{R}{\sqrt{3}}$ (2) $R\sqrt{3}$ (3) $2\sqrt{3}R$ (4) $\frac{2}{\sqrt{3}}R$

26. A cell is connected between the points A and C of a circular conductor ABCD of centre O with angle $AOC = 60^\circ$. If B_1 and B_2 are the magnitudes of the magnetic fields at O due to the currents in ABC and ADC respectively, the ratio $\frac{B_1}{B_2}$ is :-



- (1) 0.2 (2) 6 (3) 1 (4) 5

27. What will be the resultant magnetic field at origin due to four infinite length wires. If each wire produces magnetic field 'B' at origin :-



- (1) $4B$ (2) $\sqrt{2}B$ (3) $2\sqrt{2}B$ (4) Zero

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28. A circular coil of radius R carries an electric current. The magnetic field due to the coil at a point on the axis of the coil located at a distance r from the centre of the coil, ($r \gg R$), varies as :-

- (1) $1/r$ (2) $1/r^3$ (3) $1/r^2$ (4) $1/r^{3/2}$

29. During charging & discharging of a capacitor :-

- (1) Current flows in the circuit, which is constant during charging or discharging duration.
 (2) No current flows in the circuit
 (3) Current flows in the circuit and is varying with time
 (4) During charging current is constant but while discharging current is variable

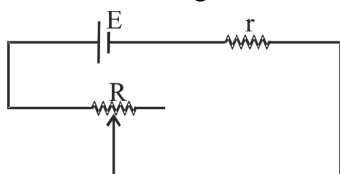
30. A galvanometer of resistance 50Ω is to be able to read up to 2mA . It is converted into an ammeter by connecting a shunt resistance of 2.5Ω then current required to produce full scale deflection in ammeter is :-

- (1) 10.5 mA (2) 21 mA
 (3) 42 mA (4) 6.3 mA

31. Potential difference ' V ' is measured across a cell of emf ' E ' then :-

- (1) E is always greater than V .
 (2) V is always greater than E .
 (3) V may be greater than E .
 (4) E and V are always equal.

32. A battery of e.m.f. E and internal resistance r is connected to a variable resistor R as shown here. Which one of the following is true



- (1) Potential difference across the terminals of the battery is maximum when $R = r$
 (2) Power delivered to the resistor is maximum when $R = r$
 (3) Current in the circuit is maximum when $R = r$
 (4) Current in the circuit is maximum when $R \gg r$

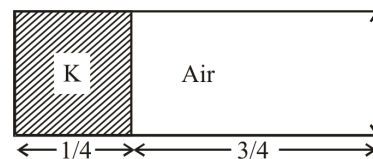
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PHASE - MEA & MEPS

33. A parallel plate capacitor of capacitance C is charged to a potential difference V by connecting it to a battery. Another capacitor of capacitance $2C$ is similarly charged to a potential difference $2V$. The charging battery is then disconnected and the capacitors are connected in parallel to each other in such a way that the positive terminal of one is connected to the negative terminal of the other. The final energy of the configuration is

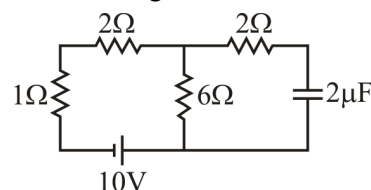
- (1) zero (2) $\frac{3}{2} CV^2$
 (3) $\frac{25}{6} CV^2$ (4) $\frac{9}{2} CV^2$

34. A parallel plate capacitor with air as the dielectric has capacitance C . A slab of dielectric constant K and having the same thickness as the separation between the plates, is introduced so as to fill one-fourth of the capacitor as shown in the figure. The new capacitance will be :



- (1) $(K+3)\frac{C}{4}$ (2) $(K+2)\frac{C}{4}$
 (3) $(K+1)\frac{C}{4}$ (4) $\frac{KC}{4}$

35. Time constant of the given circuit is :-



- (1) $2\mu\text{S}$ (2) $4\mu\text{S}$ (3) $6\mu\text{S}$ (4) $8\mu\text{S}$

SECTION-B (PHYSICS)

36. The fundamental frequency of a closed pipe is 220 Hz . If $\frac{1}{4}$ of the pipe is filled with water, the frequency of the first overtone of the pipe now is :

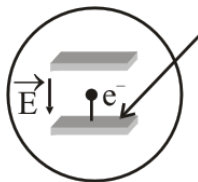
- (1) 220 Hz (2) 440 Hz
 (3) 880 Hz (4) 1760 Hz

37. Aluminium rod (Al) of length 20 cm is hinged at midpoint and longitudinal wave is generated in rod using Raisin cloth and Rod is oscillating in fundamental mode

Given $\rho_{Al} = 1600 \text{ kg/m}^3$

young's modulus for Al $= 4 \times 10^{10} \text{ N/m}^2$
frequency of fundamental mode is ?

- (1) 4 KHz (2) 6.25 KHz
(3) 25 KHz (4) 12.5 KHz
38. The momentum of a photon in an X-ray beam of 10^{-10} metre wavelength is :-
- (1) $1.5 \times 10^{-23} \text{ kg-m/sec}$
(2) $6.6 \times 10^{-24} \text{ kg-m/sec}$
(3) $6.6 \times 10^{-44} \text{ kg-m/sec}$
(4) $2.2 \times 10^{-52} \text{ kg-m/sec}$
39. When radiation of wavelength λ is incident on a metallic surface, the stopping potential is 4.8 volts. If the same surface is illuminated with radiation of double the wavelength then the stopping potential becomes 1.6 volts. Then the threshold wavelength for the surface is :-
- (1) 2λ (2) 4λ (3) 6λ (4) 8λ
40. The collector plate in an experiment on photoelectric effect is kept vertically above the emitter plate. Light source is put on and a saturation photo current is recorded. An electric field is switched on which has a vertically downward direction. Then :-



- (1) The photo current will increase
(2) The kinetic energy of the electrons will increase
(3) The stopping potential will decrease
(4) The threshold wavelength will increase

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PHASE - MEA & MEPS

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41. Match list I and II select the correct answer for option (A) (B) (C) and (D) respectively

(A) spring constant	(1) $M^1 L^2 T^{-2}$
(B) pressure	(2) $M^0 L^0 T^{-1}$
(C) frequency	(3) $M^1 L^0 T^{-2}$
(D) work	(4) $M^1 L^{-1} T^{-2}$

- (1) 3 4 2 1 (2) 4 3 1 2
(3) 4 3 2 1 (4) 3 4 1 2

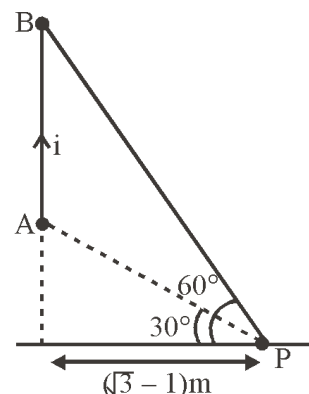
42. The least count of a stop watch is $\frac{1}{5}$ s. The time of 20 oscillations of a pendulum is measured as 25s. The percentage error in the measurement of time will be :-

- (1) 0.1% (2) 0.8% (3) 1.8% (4) 8%

43. If unit of time is doubled then how many times the unit of length must be changed so that acceleration due to gravity does not change ?

- (1) 4 (2) 2 (3) 8 (4) $\sqrt{2}$

44. A straight wire current element is carrying current 100A, as shown in figure. The magnitude of magnetic field at point P which is at perpendicular distance $(\sqrt{3} - 1)\text{m}$ from the current element if end A and end B of the element subtend angle 30° and 60° at point P, as shown, is :-



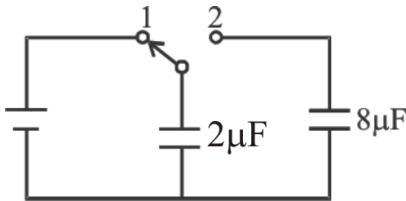
- (1) $5 \times 10^{-6} \text{ T}$
(2) $2.5 \times 10^{-6} \text{ T}$
(3) $2.5 \times 10^{-5} \text{ T}$
(4) $8 \times 10^5 \text{ T}$

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45. An equilateral triangular loop of wire of side ℓ carries a current i . The magnetic field produced at the circum center of the loop is :-

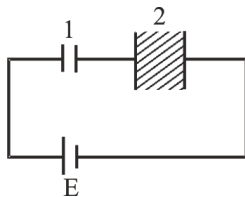
- (1) $\frac{\mu_0}{4\pi} \frac{3\sqrt{3}i}{\ell}$
- (2) $\frac{\mu_0}{4\pi} \frac{9i}{\ell}$
- (3) $\frac{\mu_0}{4\pi} \frac{18i}{\ell}$
- (4) $\frac{\mu_0}{4\pi} \frac{6i}{\ell}$

46. A $2\mu\text{F}$ capacitor is charged as shown in the figure. The percentage of its stored energy dissipated after the switch S is turned to position 2 is



- (1) 0%
- (2) 20%
- (3) 75%
- (4) 80%

47. Two identical capacitors 1 and 2 are connected in series to a battery as shown in figure. Capacitor 2 contains a dielectric slab of dielectric constant k as shown. Q_1 and Q_2 are the charges stored in the capacitors. Now the dielectric slab is removed and the corresponding charge are Q_1' and Q_2' . Then:-



- (1) $\frac{Q_1'}{Q_1} = \frac{k+1}{k}$
- (2) $\frac{Q_2'}{Q_2} = \frac{k+1}{2}$
- (3) $\frac{Q_2'}{Q_2} = \frac{k+1}{2k}$
- (4) $\frac{Q_1'}{Q_1} = \frac{k}{2}$

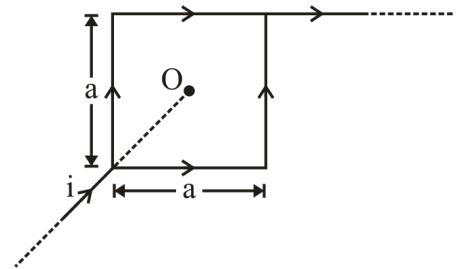
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PHASE - MEA & MEPS

48. A long straight wire along the z -axis carries a current I in the negative z direction. The magnetic vector field $\vec{B}$ at a point having coordinates (x, y) in the $z = 0$ plane is :-

- (1) $\frac{\mu_0 I (y\hat{i} - x\hat{j})}{2\pi(x^2 + y^2)}$
- (2) $\frac{\mu_0 I (x\hat{i} + y\hat{j})}{2\pi(x^2 + y^2)}$
- (3) $\frac{\mu_0 I (x\hat{j} - y\hat{i})}{2\pi(x^2 + y^2)}$
- (4) $\frac{\mu_0 I (x\hat{j} + y\hat{i})}{2\pi(x^2 + y^2)}$

49. Magnetic field at centre 'O' due to given current distribution.



- (1) $\frac{\mu_0 i}{2\sqrt{2}\pi a} (\sqrt{2} - 1) \otimes$
- (2) $\frac{\mu_0 i}{2\sqrt{2}\pi a} (\sqrt{2} - 1) \odot$
- (3) $\frac{2\sqrt{2}\mu_0 i}{\pi a} \left(1 - \frac{1}{\sqrt{2}}\right) \otimes$
- (4) Zero

50. A 120 V heater has two settings, 850 W and 1250 W. then,

- (1) resistance of 850 W is greater
- (2) resistance of 1250 W is greater
- (3) both have same resistances
- (4) data is insufficient for reaching any conclusion

Topic : p-BLOCK (HALOGEN, INERT GAS), HYDROGEN, s, d & f-BLOCK, METALLURGY, ISOMERISM (FULL), GOC-I (TILL TAUGHT), [BACK UNIT] : CHEMICAL BONDING

SECTION-A (CHEMISTRY)

51. Which one of the following oxides is expected to exhibit paramagnetic behaviour:-

- (1) CO_2 (2) ClO_2 (3) BaO (4) TiO_2

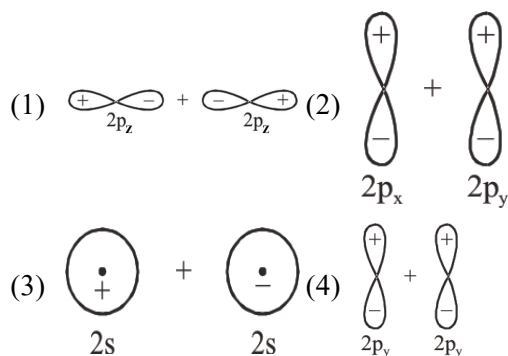
52. In ICl_2^+ , ICl_2^- and ICl_4^- sum of the bond pairs and lone pairs on each iodine atom are respectively.

- (1) 2, 2, and 4 (2) 2, 3 and 2
(3) 4, 5 and 4 (4) 4, 5 and 6

53. Dipole moment of NF_3 is less than that of :

- (1) NH_3 (2) CO_2 (3) BF_3 (4) CCl_4

54. If internuclear axis is x-axis then which of the following overlapping results in formation of bonding molecular orbital ?



55. Highest stable Mn fluoride is MnF_4 where as highest Mn oxide is Mn_2O_7 due to—

- (1) O atom is smaller than F.
(2) Oxygen has ability to form multiple bonds
(3) Mn^{7+} does not exist
(4) F can not-stabilise higher oxidation states.

56. Lanthanoid contraction is caused due to

- (1) the same effective nuclear charge from Ce to Lu
(2) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
(3) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
(4) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge

57. Which metal easily gives H_2 gas when react with NaOH ?

- (1) Na (2) Zn (3) K (4) Rb

58. Plaster of Paris, a white powder, is—

- (1) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (2) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
(3) CaSO_4 (4) $\text{CaSO}_4 \cdot \text{H}_2\text{O}$

59. Select correct statement :-

- (1) In ellingham diagram, metal 'M' which is present below any other metal oxide $\text{M}'\text{O}$ then metal 'M' is a good reducing agent for all $\text{M}'\text{O}$
(2) Due to basic nature of oxides of alkaline earth elements, they combine with atmospheric acidic oxides giving salts
(3) Both (1) and (2) are correct
(4) Th is lanthanoid element

60. $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{X} + \text{HCl}$ X Will be $\rightarrow$

- (1) HOCl (2) HClO_2 (3) HClO_3 (4) HClO_4

61. Alkali metals when exposed to moist air tarnish quickly due to the

- (1) Formation of their hydroxides
(2) Formation of their carbonates
(3) Formation of their oxides
(4) All the above

62. Which among the following is the most reactive ?

- (1) I_2 (2) ICl (3) Cl_2 (4) Br_2

63. The method of zone refining of metals is based on principle of :

- (1) Greater mobility of the pure metal than that of impurity
(2) Higher melting point of the impurity than that of the pure metal
(3) Greater noble character of the solid metal than that of impurity
(4) Greater solubility of the impurity in molten state than in the solid

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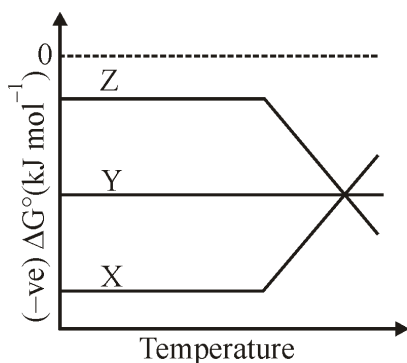
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64. **Assertion** : Tungsten has a very high melting point.

Reason : Tungsten is a covalent compound.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false.
- (3) **Assertion** is false but **Reason** is true.
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

65. In the following Ellingham diagram X, Y and Z represents graph for metal oxides. Select the correct option.



- (1) Y will reduce oxide Z
- (2) Y will reduce oxide X
- (3) Z will reduce oxide X
- (4) Z will reduce oxide Y

Column-I		Column-II	
(A)	BrF_3	(P)	One angle $\leq 90^\circ$
(B)	TeF_5^-	(Q)	Central atom is sp^3d^2 hybridised
(C)	IF_7	(R)	Non planar
(D)	XeF_4	(S)	Polar

- (1) (A) - P,S; (B) - P,Q,RS; (C) - P,R; (D) - P,Q
- (2) (A) - P,Q; (B) - P,RS; (C) - P,S; (D) - P,R
- (3) (A) - P,Q,RS; (B) - P,Q; (C) - P,S; (D) - P,R
- (4) (A) - P,R; (B) - P,Q; (C) - P,S; (D) - P,Q,RS

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67. Maximum bond energy is in :

- (1) F_2
- (2) N_2
- (3) O_2
- (4) All molecule has equal Bond energy

68. The order of strength of hydrogen bonds is :

- (1) $\text{ClH} \cdots \text{Cl} > \text{NH} \cdots \text{N} > \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$
- (2) $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} < \text{OH} \cdots \text{O} < \text{FH} \cdots \text{F}$
- (3) $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} > \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$
- (4) $\text{ClH} \cdots \text{Cl} < \text{NH} \cdots \text{N} < \text{OH} \cdots \text{O} > \text{FH} \cdots \text{F}$

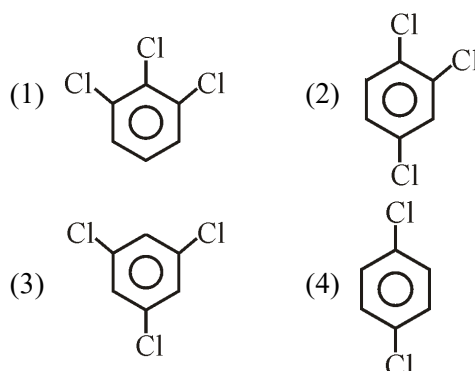
69. The strongest force among the following is

- (1) London force
- (2) Ion-dipole interaction
- (3) Dipole-induced dipole interaction
- (4) Dipole-dipole interaction

70. Xe can be liquified due to "_____" with in molecules :

- (1) ion-dipole interaction
- (2) dipole-dipole interaction
- (3) dipole-induced dipole interaction
- (4) instantaneous dipole-induced dipole interaction

71. Which has maximum dipole moment ?



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72. Which of the following option is incorrect ?

(I) All C–O bonds in CO_3^{2-} are equal but not in H_2CO_3 .

(II) All C–O bonds in HCO_2^- are equal but not in HCO_2H

(III) C–O bond length in HCO_2^- is longer than C–O bond length in CO_3^{2-}

(IV) C–O bond length in HCO_2^- and C–O bond length in CO_3^{2-} are equal.

(1) III & IV (2) II & III

(3) II & IV (4) I & IV

73. Which of the following would exhibit cis-trans isomerism :-

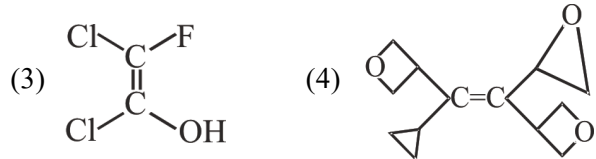
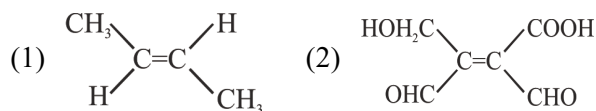
(1) $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$

(2) $\text{ClCH}=\text{CHCl}$

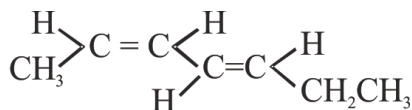
(3) $\text{ClCH}=\text{CCl}_2$

(4) $\text{CH}_2=\text{CH}-\text{COOH}$

74. Which of the following has "Cis" as well as "E" configuration?



75. What is the IUPAC name for the following compound?



(1) cis, trans-2, 4-heptadiene

(2) cis, cis-2, 4-heptadiene

(3) trans, trans-2, 4-heptadiene

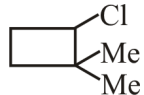
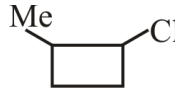
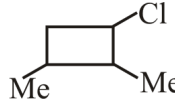
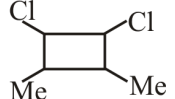
(4) 2E, 4E-2, 4-heptadiene

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76. Match the column

Column - I		Column - II	
(a)		(p)	Number of chiral center = 1
(b)		(q)	Number of chiral center = 2
(c)		(r)	Number of chiral center = 3
(d)		(s)	Number of chiral center = 4

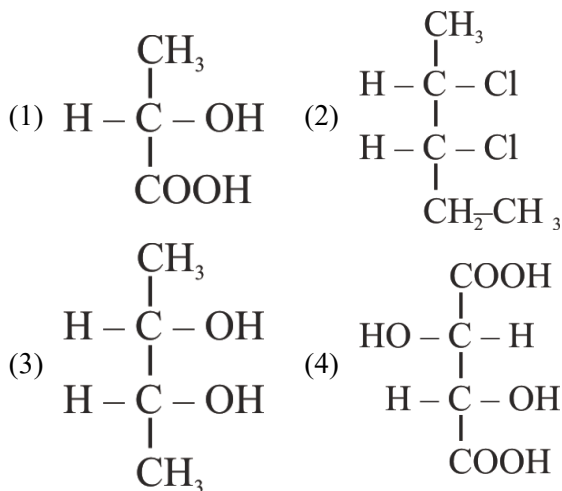
(1) $a \rightarrow p, b \rightarrow q, c \rightarrow r, d \rightarrow s$

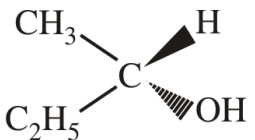
(2) $a \rightarrow q, b \rightarrow r, c \rightarrow p, d \rightarrow s$

(3) $a \rightarrow p, b \rightarrow s, c \rightarrow r, d \rightarrow q$

(4) $a \rightarrow p, b \rightarrow q, c \rightarrow s, d \rightarrow r$

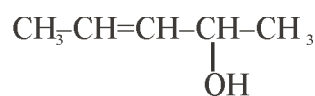
77. Which of the following compound although have chiral center but optically inactive ?



78.  is :-

(1) R- (2) S- (3) E- (4) Z-

79. Number of stereoisomers of the given compound is :-



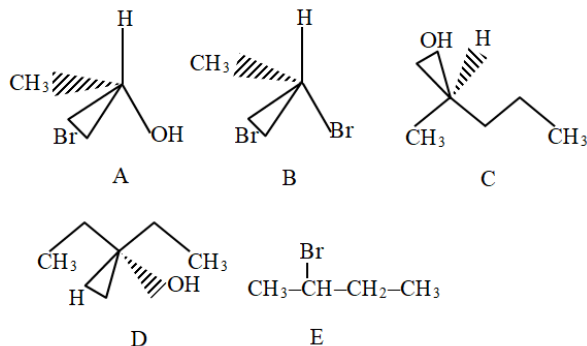
(1) 2 (2) 3 (3) 4 (4) 6

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80. Which is an electrophile :-

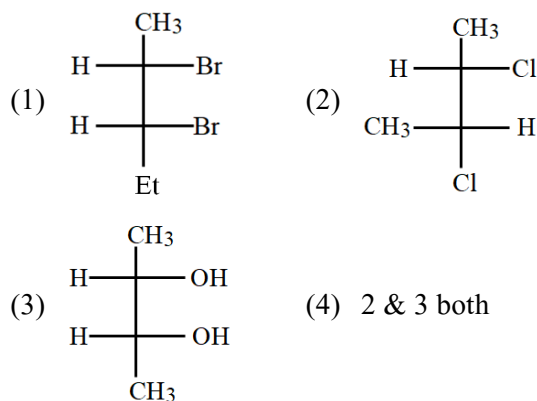
- (1) NH_4^+ (2) SO_3
 (3) $\text{CH}\equiv\text{CH}$ (4) H_3O^+

81. Identify chiral molecule

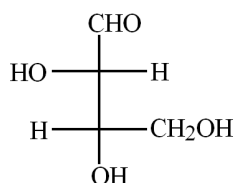


- (1) A, C, E (2) A, C, D, E
 (3) A, E (4) All

82. Which of the following is meso compound ?



83. Find correct configuration of given molecule



- (1) Erythro, D (2) Threo, D
 (3) Erythro, L (4) Threo, L

84. Which is incorrect order.

- (1) $\text{CH}_2=\text{CH}_2 > \text{CH}_3\text{CH}=\text{CH}_2$ (Boiling point)
 (2) $\text{CH}_2=\text{CH}_2 > \text{CH}_3\text{CH}=\text{CH}_2$ (Melting point)
 (3) $\text{CH}_2=\text{CH}_2 > \text{CH}_3\text{CH}=\text{CH}_2$ (Solubility in water)
 (4) $\text{CH}_2=\text{CH}_2 > \text{CH}_3\text{CH}=\text{CH}_2$ (Dipole)

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85. Oxime of which compound can show geometrical isomerism ?

- (1) HCHO (2) CH_3COCH_3
 (3) Acetophenone (4) Benzophenone

SECTION-B (CHEMISTRY)

86. Match the compounds given in column I with the shape given in column II and mark the correct option.

	Column-I	Column-II	
(a)	XeF_6	(i)	Distorted octahedral
(b)	XeO_3	(ii)	Square planar
(c)	XeOF_4	(iii)	pyramidal
(d)	XeF_4	(iv)	Square pyramidal

Code :-

- (a) (b) (c) (d)
 (1) (i) (iii) (iv) (ii)
 (2) (i) (ii) (iv) (iii)
 (3) (iv) (iii) (i) (ii)
 (4) (iv) (i) (ii) (iii)

87. The correct electronic configuration of Gd^{+3} ion is

- (1) $[\text{Rn}] 7s^2 6d^1 5f^7$ (2) $[\text{Xe}] 6s^2 5d^0 4f^7$
 (3) $[\text{Rn}] 5f^7$ (4) $[\text{Xe}] 4f^7$

88. The pair of elements which on combination are most likely to form an ionic compound is

- (1) Na and Ca (2) K and O
 (3) O and Cl (4) Al and I

89. $\text{H}_2\text{O} > \text{H}_2\text{O}_2$ is correct for :-

- (1) Number of H bond (2) Acidic character
 (3) H bond strength (4) Boiling point

90. The function of florspar (CaF_2) in electrolytic reduction of alumina is :-

- (1) as a catalyst
 (2) To lower the melting point and to increase the conductivity
 (3) To decrease the rate of oxidation of carbon at anode
 (4) all of these

91. **Assertion :** Lead, tin & Bismuth are purified by liquation method.

Reason: Lead, tin and Bismuth have low melting point as compared to impurities.

- (1) Both Assertion and Reason are correct and the Reason is correct explanation of the Assertion
- (2) Both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion
- (3) Assertion is correct but Reason is incorrect
- (4) Both Assertion and Reason are incorrect.

92. H_2O_2 does not act as

- (1) Reducing agent
- (2) Oxidising agent
- (3) Dehydrating agent
- (4) Bleaching agent

93. Argon is used in arc welding because of its -

- (1) low reactivity with metal
- (2) ability of lower the melting point of metal
- (3) flammability
- (4) high calorific value

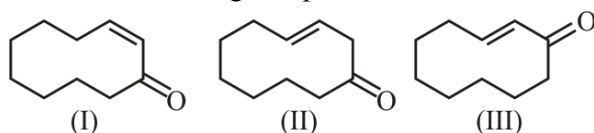
94. During estimation of oxalic acid Vs $KMnO_4$, self indicator is

- (1) $KMnO_4$
- (2) oxalic acid
- (3) K_2SO_4
- (4) $MnSO_4$

95. Bond order of Be_2 is:

- (1) 1
- (2) 2
- (3) 3
- (4) 0

96. Which of the following statement is not true about the following compounds :-

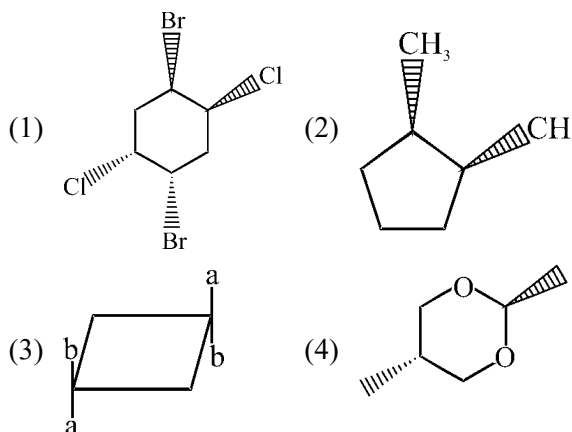


- (1) I & III are identical
- (2) I & III are geometrical isomers
- (3) I & II are structural isomers
- (4) II & III are structural isomers

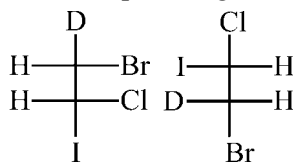
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PHASE - MEA & MEPS

97. Which have centre of symmetry (cos) and plane of symmetry (pos) both :-

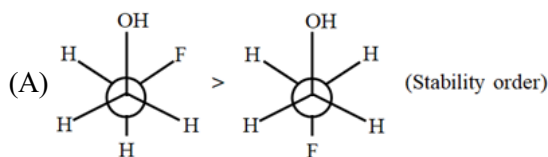


98. Two compounds given below are :



- (1) Enantiomer
- (2) Identical
- (3) Meso compound
- (4) Diastereomers

99. Consider following statements -



(B) $C_5H_{12}O$ has 12 structural isomers.

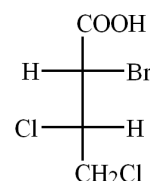
(C) are positional isomer

(D) $CH_3-CH_2-C(=O)-CH_2-CH_3$ & $CH_3-C(=O)-CH_2-CH_2-CH_3$ are metamer

Correct statement among above are

- (1) A, B, C
- (2) A, D
- (3) B, C, D
- (4) All

100. Absolute configuration of given molecule



- (1) 2R, 3S
- (2) 2S, 3S
- (3) 3R, 2S
- (4) 2R, 3R.

Topic : PRINCIPLES OF INHERITANCE AND VARIATIONS, COMPLETE PHOTOSYNTHESIS (EXCEPT HISTORY), 4 FAMILIES AND ANATOMY (EXCLUDING SECONDARY GROWTH), [BACK UNIT] : HUMAN HEALTH AND DISEASE

SECTION-A (BIOLOGY-I)

101. During blackman reaction of photosynthesis :-

- (1) Water is oxidised and NADP get reduced
- (2) Both CO_2 and water get reduced
- (3) Water get reduced
- (4) CO_2 get reduced and NADPH.H^+ get oxidised

102. Radioisotopic technique was used to :-

- (1) Find out the absorption spectrum of chlorophyll pigments
- (2) Study photorespiration in animals.
- (3) Establish the fact that the oxygen evolved during photosynthesis comes from CO_2
- (4) Trace the path of carbon in photosynthesis

103. $2\text{H}_2\text{A} + \text{CO}_2 \rightarrow 2\text{A} + \text{CH}_2\text{O} + \text{H}_2\text{O}$

If this reaction is occur in cyanobacteria then what is A :-

- (1) Sulphur
- (2) Water
- (3) Nitrogen
- (4) Oxygen

104. Absorption of radiant energy causes :-

- (1) Reduction of chlorophyll
- (2) Oxidation of chlorophyll
- (3) Absorption of CO_2
- (4) Fixation of N_2

105. Ferredoxin is a component of -

- (1) RuBisCO
- (2) PS I
- (3) PS II
- (4) PEPcase

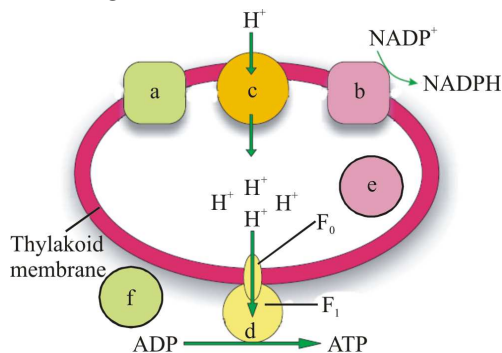
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PHASE - MEA & MEPS

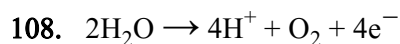
106. Which of the following components are required for chemiosmosis ?

- (1) A membrane, a proton pump, an electron gradient and ATPase
- (2) $\text{F}_0\text{--F}_1$ particle, ATP, a membrane and proton gradient
- (3) A proton pump, a membrane, a proton gradient and ATPsynthase
- (4) A proton pump, a membrane, a proton gradient and PS-II

107. Recognise the figure and find out the correct matching a, b, c, d, e and f :-



- (1) a-ATPsynthase, b-photosystem I, c-photosystem II, e-stroma, f-lumen, d-cytochrome b and f
- (2) d-ATPsynthase, a-photosystem I, b-photosystem II, f-stroma, e-lumen, c-cytochrome b and f
- (3) d-ATPsynthase, b-photosystem I, a-photosystem II, f-stroma, e-lumen, c-cytochrome b and f
- (4) d-ATPsynthase, a-photosystem I, b-photosystem II, e-stroma, f-lumen, c-cytochrome b and f



Consider the following with respect to above equation.

- Site of this reaction is stroma side of thylakoid membrane
- Associated with PS-II
- Take place in both mitochondria and chloroplast
- Source of proton in both cyclic and non cyclic photophosphorylation

How many are not related ?

- (1) 4 (2) 3 (3) 2 (4) 1

109. At which step of Calvin cycle the energy products of non-cyclic process are used ?

- Regeneration
- Carboxylation
- Reduction
- Transamination

110. (i) using a prism he split light into its spectral components and then illuminated a green alga (ii) placed in a suspension of (iii). The bacteria were used to detect the sites of O_2 evolution. He observed that the bacteria accumulated mainly in the region of blue and red regions of the split spectrum. A first (iv) of photosynthesis was thus described.

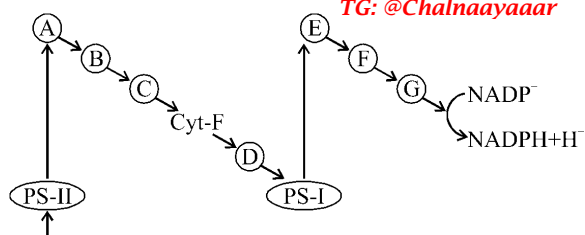
- (i)-T.W. Engelmann, (ii)-Chlorella, (iii)-Anaerobic bacteria, (iv)-Action spectrum.
- (i)-White man, (ii)-Cladophora, (iii)-Aerobic bacteria, (iv)-Absorption spectrum
- (i) Black man, (ii)-Scenedesmus, (iii)-Anaerobic bacteria, (iv)-Absorption spectrum
- (i)-T.W. Engelmann, (ii)-Cladophora, (iii)-Aerobic bacteria, (iv)-Action spectrum

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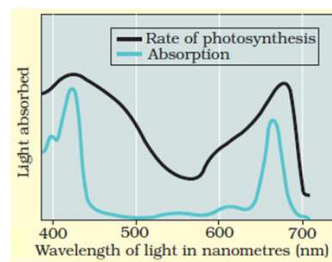
111.



In the above diagram, which one is copper containing mobile e^- carrier.

- (1) A (2) B (3) C (4) D

112. The given graph represents action spectrum of photosynthesis and absorption spectrum of chlorophyll -a. Select the correct statement regarding this graph.



- This graph shows that photosynthesis occurs only in blue and red regions of visible spectrum.
- It shows that chlorophyll-a absorbs entire light energy.
- It can be concluded that chlorophyll-a is chief pigment involved in photosynthesis.
- It shows complete one to one overlap between actions spectrum and absorption spectrum of chlorophyll-a.

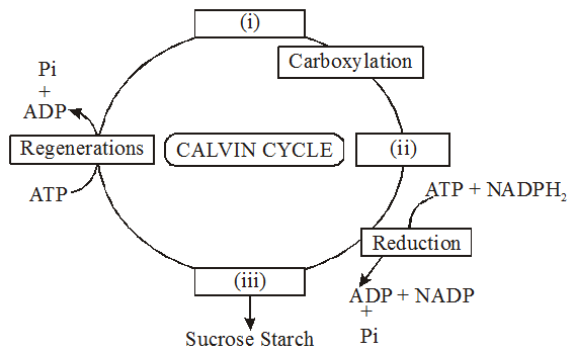
113. Match the column :-

	Column-I		Column-II
(a)	$\text{Mn}^{++}, \text{Cl}^-$	(i)	First stable product of C_3 -cycle
(b)	OAA	(ii)	Primary acceptor of electron in non cyclic electron transport chain
(c)	Pheophytin	(iii)	Carboxylation of RUBP
(d)	RuBisCO	(iv)	First stable product of C_4 -cycle
(e)	3-PGA	(v)	Photolysis of water

- a-v, b-i, c-ii, d-iii, e-iv
- a-iii, b-iv, c-i, d-v, e-ii
- a-ii, b-i, c-iii, d-v, e-iv
- a-v, b-iv, c-ii, d-iii, e-i

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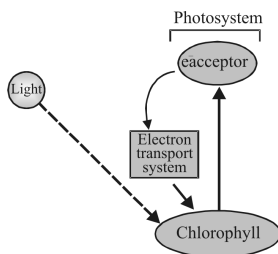
114. Choose the correct combinations of labelling the molecules involved in the Calvin cycle :-



- (1) (i) RuBP (ii) Triose phosphate (iii) PGA
 (2) (i) PGA (ii) RuBP (iii) Triose phosphate
 (3) (i) PGA (ii) Triose phosphate (iii) RuBP
 (4) (i) RuBP (ii) PGA (iii) Triose phosphate
115. How many components in the list given below participate both in cyclic and non-cyclic photophosphorylation ?
 Pheophytin, Plastoquinone, cytochrome b_6 , Ferredoxin, Ubiquinone, cytochrome f , plastocyanin, OEC

- (1) Six (2) Three (3) Four (4) Five

116.



- a-Photophosphorylation occur
 b-PS-II involved
 c-Enzyme NADP reductase is used
 d-Fe-S as primary electron acceptor
 e-Takes place in stroma lamellae
 f-Cannot operate above 680 nm of light
 g-Provides additional ATP. required in dark reaction.

h-No photolysis of H_2O .

How many of the above statements apply to the above shown process ?

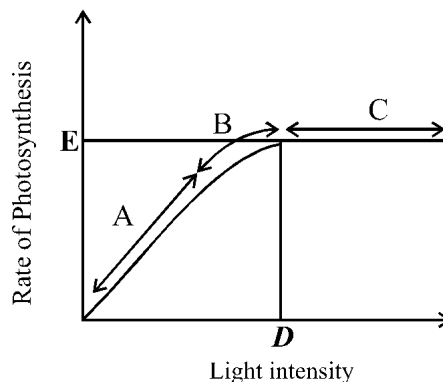
- (1) a, d, e, g, h (2) a, c, d, e, g
 (3) b, d, f, g, h (4) a, b, e, f, h

117. **Assertion** : Water stress can indirectly effect photosynthesis.

Reason : Water stress leads to closure of stomata and prevents gaseous exchange.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
 (2) **Assertion** is true but **Reason** is false.
 (3) **Assertion** is false but **Reason** is true
 (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**

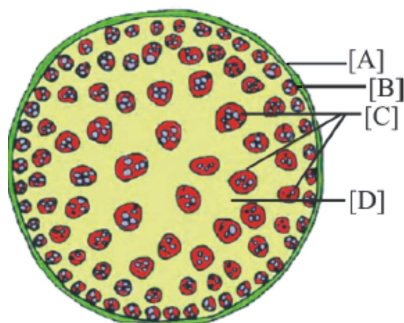
118. How many statements are correct regarding the relationship of rate of photosynthesis with intensity of light as depicted in the graph given below :



- (i) At low light intensity, chlorophyll act as a limiting factor
 (ii) In segment 'C' of the curve, light intensity is not a limiting factor
 (iii) Light compensation point is observed in segment 'C' of the curve.
 (iv) At high light intensity, rate of photosynthesis is saturated as light is no more utilized in photosynthesis

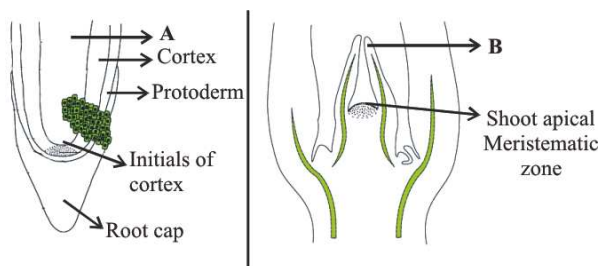
- (1) Four
 (2) Two
 (3) One
 (4) Three

119. Given below is the diagrammatic view of transverse section of monocot stem. Identify the parts labelled A, B, C and D and select the right option about them



	Part A	Part B	Part C	Part D
(1)	Epidermis	Hypodermis	Ground tissue	Vascular bundles
(2)	Epidermis	Hypodermis	Vascular bundles	Ground tissue
(3)	Hypodermis	Epidermis	Ground tissue	Vascular bundles
(4)	Hypodermis	Ground tissue	Epidermis	Vascular bundles

120. Identify A & B in given diagrams :-



	A	B
(1)	Root apex	Leaf primordium
(2)	Stele (Central cylinder)	Leaf
(3)	Stele (Central cylinder)	Leaf primordium
(4)	Endodermis	Apical bud

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121. Which of the following statement is not correct :-

- (1) Study of the internal structure of plant is known as plant anatomy.
- (2) Plant tissues are classified on the basis of their capability of division.
- (3) All the lateral meristems are secondary meristems
- (4) All the secondary meristems are lateral meristems

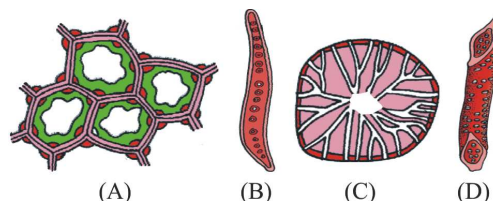
122. Companion cells are and maintain the in sieve tubes :-

- (1) Specialised collenchymatous cells, concentration.
- (2) Specialised phloem cells, atmospheric pressure and temperature.
- (3) Specialised parenchymatous cells, pressure gradient.
- (4) Both (2) and (3)

123. In grasses, certain epidermis cells along the veins modify themselves into large, empty, colourless cells These are called :-

- (1) Lateral, cork cells
- (2) Lower, epithem cells
- (3) Adaxial, bulliform cells
- (4) Abaxial, epithem cells

124. Given below is the four (A, B, C and D) sketches, identify them :-



	A	B	C	D
1	Parenchyma	Vessel	Sclereid	Tracheid
2	Collenchyma	Tracheid	Sclereid	Vessel
3	Collenchyma	Parenchyma	Tracheid	Vessel
4	Sclereid	Tracheid	Collenchyma	Sieve tube element

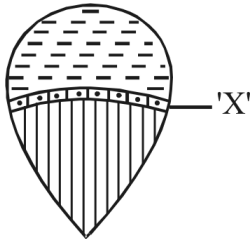
125. Living mechanical tissue with localized deposition of pectin at corners is -

- (1) Parenchyma (2) Collenchyma
(3) Chlorenchyma (4) Aerenchyma

126. Which of the following is not a part of the ground tissue system ?

- (1) General cortex (2) Endodermis
(3) Xylem (4) Pericycle

127. In the given figure labelled region 'X' is :-

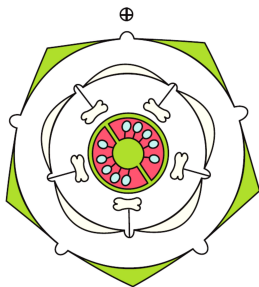


- (1) Primary phloem
(2) Secondary phloem
(3) Intra fascicular cambium
(4) Primary xylem

128. Conjoint vascular bundles are common in :-

- (1) Root (2) Stem
(3) Leaf (4) Both (2) & (3)

129. For the given following floral diagram correct floral formula is :-



- (1) $\oplus \overline{\text{K}}_{(5)} \overline{\text{C}}_{(5)} \overline{\text{A}}_5 \underline{\text{G}}_{(2)}$
(2) $\oplus \overline{\text{K}}_{(5)} \overline{\text{C}}_{(5)} \overline{\text{A}}_{(5)} \underline{\text{G}}_{(2)}$
(3) $\oplus \overline{\text{K}}_{(5)} \overline{\text{C}}_{(5)} \overline{\text{A}}_5 \underline{\text{G}}_{(2)}$
(4) $\oplus \overline{\text{K}}_{2+2} \overline{\text{C}}_{4x} \overline{\text{A}}_{2+4} \underline{\text{G}}_{(2)}$

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PHASE - MEA & MEPS

130. Monocarpellary gynoecium and diadelphous androecium and marginal placentation occur in :-

- (1) Brassicaceae (2) Solanaceae
(3) Liliaceae (4) Fabaceae

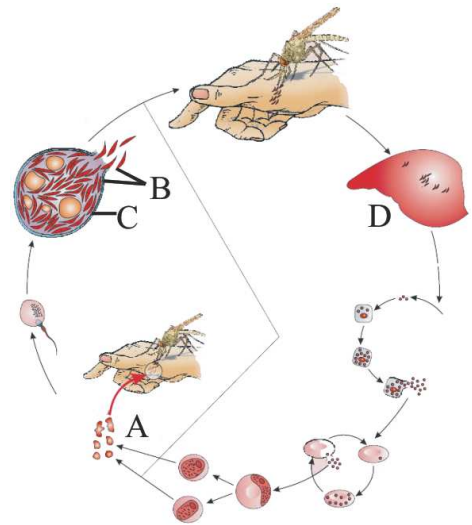
131. Pentamerous, actinomorphic flower, bicarpellary syncarpous ovary with oblique septa and fruit is berry or capsule are characteristic feature of :-

- (1) Fabaceae (2) Solanaceae
(3) Cruciferae (4) Liliaceae

132. Which of the following plant is not related to the family Fabaceae ?

- (1) Sunnhemp (2) *Indigofera* (Neel)
(3) *Aloe vera* (4) Lupin

133. Identify the likely A, B, C, D in the following life cycle



	(A)	(B)	(C)	(D)
(1)	Sporozoite	Gametocyte	Salivary gland	Liver
(2)	Gametocyte	Sporozoite	Liver	Salivary gland
(3)	Sporozoite	Gametocyte	Liver	Salivary gland
(4)	Gametocyte	Sporozoite	Salivary gland	Liver

134. Match the columns :

	Column-I		Column-II
A.	Malignant tumors	i.	Hallucinogenic
B.	MRI	ii.	Metastasis
C.	Morphine	iii.	Non-ionising radiations
D.	Atropa belladonna	iv.	Depressant

(1) A-i, B-ii, C-iii, D-iv

(2) A-ii, B-iii, C-iv, D-i

(3) A-iii, B-i, C-ii, D-iv

(4) A-iv, B-ii, C-i, D-iii

135. 'Infected cell can survive while viruses are being replicated and released'. In case of AIDS given statement is true for which cell ?

- (1) T-Helper (2) T-Killer
(3) Macrophage (4) B-lymphocyte

SECTION-B (BIOLOGY-I)

136. Chlorophyll is :-

- (1) Fe, porphyrin with hydrophilic head and hydrophobic tail
(2) Mg, porphyrin with hydrophobic head and hydrophilic tail
(3) Mg, porphyrin with hydrophilic head and hydrophobic tail
(4) Fe, porphyrin with hydrophobic head and hydrophilic tail.

137. Select the incorrect statement about non-cyclic photophosphorylation :-

- (1) In PS-II the reaction centre Chl-a absorbs 680 nm wavelength of red light
(2) Electrons in the reaction centre of PS-I are excited when they receive red light of wavelength 700 nm
(3) NADP^+ is reduced to $\text{NADPH} + \text{H}^+$
(4) The excited electron does not pass on to NADP^+ but is cycled back to the PS-I complex through the ETS

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138. $\text{A} + \text{CO}_2 \xrightarrow{\text{B}} 2 \times 3 \text{PGA}$

Here A and B are respectively ?

- (1) Malic acid, PEP carboxylase.
(2) RuBisCo, RuBP
(3) RuBP, RuBisCo
(4) Glucose, Hexokinase

139. The number of photons required for the evolution of 5 molecules of O_2 in photosynthesis is :-

- (1) 10 (2) 05 (3) 40 (4) 30

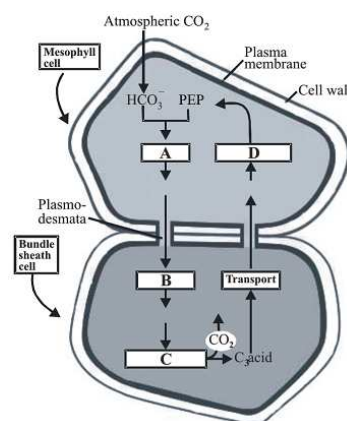
140. How many $\text{NADPH} + \text{H}^+$ are produced in cyclic photophosphorylation per $4e^-$ flow in its ETS :-

- (1) One (2) Two (3) Four (4) Zero

141. How many ATP and NADPH are used in the synthesis of two glucose molecules in the reduction step of C_3 plant respectively :-

- (1) 12 ATP + 12 NADPH
(2) 12 ATP + 18 NADPH
(3) 24 ATP + 24 NADPH
(4) 36 ATP + 24 NADPH

142. The figure below shows steps (A, B, C, D) of C_4 pathway. Select the option giving correct identification together with correct statement?



- (1) A-Fixation carried out by PEPCase enzyme
(2) B-Transport of C_3 acid
(3) C-Decarboxylation of oxaloacetate
(4) D-Regeneration with the help of malic enzyme

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143. Find out the correct match from the following table :-

	Column I	Column II	Column III
(i)	Apical Meristem	Shoot apex	Regeneration of grasses
(ii)	Intercalary Meristem	occurs between mature tissues	Forms axillary bud
(iii)	Lateral Meristem	Cylindrical	Secondary Growth

- (1) (i) only (2) (i) and (ii)
 (3) (iii) only (4) (ii) and (iii)

144. Consider the following four statements (a-d) and select the option which includes all the correct ones only.

- (a) The innermost layer of the cortex is called pericycle. It comprises a single layer of barrel shaped cells.
 (b) The tangential as well as radial walls of the endodermal cells of roots have deposition of a waxy material suberin.
 (c) Initiation of lateral roots takes place from the pericycle of roots.
 (d) Pith is large and well developed in monocot roots.

Options :-

- (1) Statements (b), (c) and (d)
 (2) Statements (a), (b)
 (3) Only statements (c), (d)
 (4) Statements (a), (c) and (d)

145. Which of the following statements is not correct ?

- (1) Tissue systems are of three types on the basis of their structure and location.
 (2) Epidermal tissue system comprises root hair, trichomes and stomata.
 (3) Epidermis is parenchymatous
 (4) Waxy layer over epidermis is composed of suberin

146. Identify the tissue on the basis of character given below :-

- (i) Cell with protoplasm
 (ii) Usually intercellular spaces are absent
 (iii) Cell with thickened cell wall at the corners
 (iv) Found in young dicot stem

- (1) Xylem (2) Phloem
 (3) Parenchyma (4) Collenchyma

147. Which of the following condition is not found in Liliaceae family ?

- (1) Exstipulate leaf (2) Axile placentation
 (3) Endospermic seeds (4) Epipetalous condition

148. The correct floral formula of pea is:-

- (1) $\oplus \text{ } \overline{\text{P}}_{3+3} \text{ } \overline{\text{A}}_{3+3} \text{ } \underline{\text{G}}_{(3)}$
 (2) $\% \text{ } \overline{\text{K}}_{(5)} \text{ } \overline{\text{C}}_{1+2+(2)} \text{ } \overline{\text{A}}_{(9)+1} \text{ } \underline{\text{G}}_1$
 (3) $\oplus \text{ } \overline{\text{K}}_{(5)} \text{ } \overline{\text{C}}_{(5)} \text{ } \overline{\text{A}}_5 \text{ } \underline{\text{G}}_{(2)}$
 (4) $\oplus \text{ } \overline{\text{K}}_{2+2} \text{ } \overline{\text{C}}_4 \text{ } \overline{\text{A}}_{2+4} \text{ } \underline{\text{G}}_{(2)}$

149. Read the following statements regarding spleen and select the correct option :

- (i) Spleen is a large bean-shaped organ which mainly contains lymphocytes and phagocytes.
 (ii) Spleen is a large reservoir of erythrocytes.
 (iii) Spleen is a primary lymphoid organ.
 (iv) Spleen acts as a filter of the blood by trapping blood-borne micro-organisms.

- (1) (i) and (ii) (2) (ii) and (iv)
 (3) (i), (ii) and (iii) (4) (i), (ii) and (iv)

150. Which of the following acts as a physiological barrier to the entry of micro organism in human body ?

- (A) Epithelium of Respiratory tract
 (B) Monocyte
 (C) Tear
 (D) Skin
 (E) Saliva
 (F) Acid in stomach
 (G) N-K-cell

- (1) E, F, G (2) B, C, E
 (3) C, E, F (4) A, B, D

Topic : PRINCIPLES OF INHERITANCE AND VARIATIONS, COMPLETE PHOTOSYNTHESIS (EXCEPT HISTORY), 4 FAMILIES AND ANATOMY (EXCLUDING SECONDARY GROWTH), [BACK UNIT] : HUMAN HEALTH AND DISEASE

SECTION-A (BIOLOGY-II)

151. Why mother's milk is considered very essential for the newborn infant :-

- (1) It has IgG antibody
- (2) It provides active immunity
- (3) Because colostrum has IgA antibody
- (4) Because IgA provides innate/inborn immunity

152.



Identify the plant and corresponding drug extracted from it :-

- (1) Erythroxylum Coca, Morphine
- (2) Claviceps Purpurea, LSD
- (3) Papaver Somniferum, Morphine
- (4) Mangifera Indica, Cocaine

153. Which of the following is correct for LSD, morphine and charas respectively?

- (1) *Claviceps*, *Papaver somniferum*, *Trichoderma*
- (2) *Claviceps*, *Cannabis*, *Cocca* plant
- (3) *Claviceps*, *Papaver somniferum*, *Cannabis*
- (4) *Papaver somniferum*, *Cleviceps*, *Cannabis*

154. Diseases are broadly grouped into infectious and non-infectious diseases. In the list given below, identify the infectious diseases.

- (i) Cancer (ii) Influenza
(iii) Allergy (iv) Small pox

- (1) (i) and (ii) (2) (ii) and (iii)
(3) (iii) and (iv) (4) (ii) and (iv)

155. The disease which is not transmitted by mosquito bite ?

- (1) Malaria (2) Filariasis
(3) Chikungunya (4) Ascariasis

156. Which of the following disease is not completely curable even if detected early and treated properly ?

- (1) Gonorrhoea (2) Syphilis
(3) Hepatitis - B (4) Trichomoniasis

157. " Don't die of ignorance" is said for ?

- (1) Typhoid (2) Pneumonia
(3) AIDS (4) Hepatitis - B

158. Which of the following non-infectious disease is a major cause of death in human being ?

- (1) Asthma
(2) AIDS
(3) Rheumatoid arthritis
(4) Cancer

159. Which of the following is a common symptom of typhoid ?

- (1) Stool with excess mucous and blood clotts
(2) Dry scaly lesions
(3) Sustained high fever (30-40°F)
(4) Loss of appetite

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PHASE - MEA & MEPS

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160. Which of the following is not an example of autosomal trisomy.
- (1) Edward syndrome
 - (2) Down syndrome
 - (3) Patau syndrome
 - (4) Kline felter syndrome
161. Which cells are not considerably decreased in number in AIDS patient ?
- (1) Helper T- cells (2) Macrophages
 - (3) CD₄ - cells (4) T₄ - cells
162. Which of the following hallucinogen is not obtained by flowering plants.
- (1) Cannabis sativa (2) Atropa belladonna
 - (3) Datura (4) L.S.D
163. The site where lymphocytes interact with antigens and proliferate to become effector cells are ?
- (1) Bone marrow and thymus
 - (2) Spleen and bone marrow
 - (3) Tonsils and lymph nodes
 - (4) Spleen and thymus
164. Which type of immunity is not responsible for graft rejection ?
- (1) Cellular immunity
 - (2) Cell mediated immunity
 - (3) 1 and 2 both
 - (4) Innate immunity
165. Toxin which is responsible for chill and high fever during malaria ?
- (1) Lymphotoxin
 - (2) Exotoxin
 - (3) Haemozoin
 - (4) Antitoxin
166. Smack is obtained by acetylation of morphine, which is extracted from the latex of ?
- (1) Erythroxylum coca (2) Cannabis sativa
 - (3) Poppy plant (4) Datura
167. Person who take drugs intravenously is more likely to acquire serious infection like ?
- (1) Cancer (2) Pneumonia
 - (3) Addiction (4) AIDS
168. The use of drugs like antihistamine adrenaline and steroids quickly reduce the symptoms of ?
- (1) Autoimmunity (2) Viral disease
 - (3) Asthma (4) Amoebiasis
169. In co-dominance the genes of an allelic pair express _____ and follow Law of _____ :-
- (1) Equally, Independent assortment
 - (2) Unequally, Dominance
 - (3) Equally, Segregation
 - (4) Equally, Dominance
170. Four couples are claiming to be the parents of a child having 'O' blood group. Which of the following couples' claim is baseless, with blood groups of male and female parents, respectively as :-
- (1) A and B
 - (2) A and O
 - (3) B and O
 - (4) AB and O
171. In case of sickle cell anaemia :-
- (1) Heterozygous is diseased
 - (2) Valine is replaced by glutamic acid
 - (3) Shape of the RBC (erythrocyte) changes
 - (4) Hb^sHb^s is normal

172. Which statement is not correct for complementary gene interaction ?

- (1) 9 : 7 ratio is obtained when F_1 hybrid are selfed.
- (2) Test cross gives 1 : 3 ratio
- (3) 2 phenotypes & 4 genotypes are produced in cross $CcPp \times ccpp$
- (4) One gene is responsible for a phenotype

173. Which of the following statement is correct about polygenic inheritance ?

- (1) In polygenic inheritance, one gene regulates the expression of many characters.
- (2) Characters which are regulated by polygenes are qualitative characters.
- (3) Besides the involvement of multiple genes, polygenic inheritance also takes into account the influence of environment.
- (4) Human skin colour is an example of qualitative character.

174. In independent assortment what actually assort independently ?

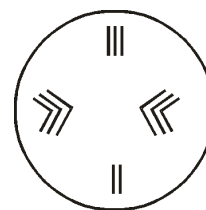
- (1) Characters
- (2) Contrasting traits
- (3) Allelic genes
- (4) Non homologous chromosomes

175. Eye colour (A) – Body colour (B) = 8%
 Eye colour (A) – Wing shape (C) = 24%
 Body colour (B) – Wing shape (C) = 16%
 Wing shape (C) – Eye shape (D) = 10%
 Body colour (B) – Eye shape (D) = 26%
 Now which of the following is correct genetic map for such values of crossing over in Drosophila genes :-

- (1)
- (2)
- (3)
- (4) Either (1) or (2)

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176. Given below is the karyotype of a Drosophila. On the basis of genic balance theory of C.B. Bridges, the sex of this fly would be :-



- (1) Super female
- (2) Super male
- (3) Inter sex
- (4) Fertile female

177. A colourblind boy has 2 sisters, one colourblind and one normal. What can be the possible nature of their parents ?

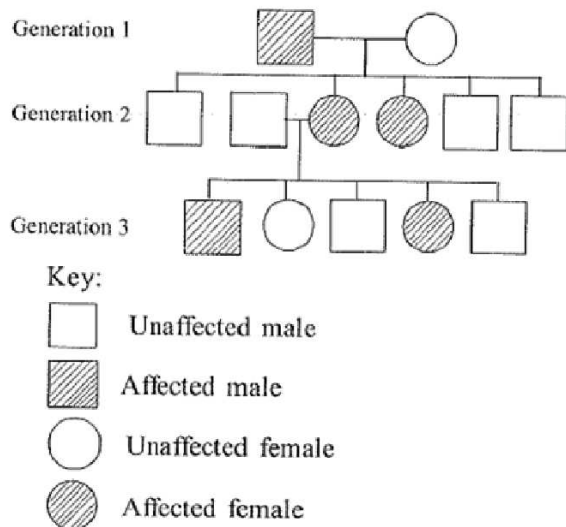
- (1) Colourblind father and colourblind mother
- (2) Normal father and colourblind mother
- (3) Colourblind father and carrier mother
- (4) Colourblind father and normal mother

178.

Phenotype	Genotype	Phenotype
50 cm	TT	50 cm
35 cm	Tt	50 cm
20 cm	tt	20 cm
A		B

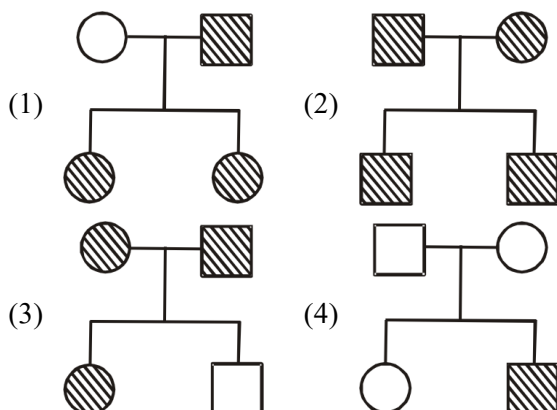
	A	B
(1)	Polygenic qualitative	Polygenic quantitative
(2)	Monogenic quantitative	Monogenic qualitative
(3)	Monogenic quantitative	Monogenic quantitative
(4)	Monogenic qualitative	Monogenic qualitative

179. Given below is a pedigree chart showing the inheritance of a certain sex-linked trait in humans

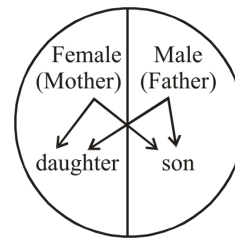


The trait shown in the above pedigree chart is

- (1) Dominant X-linked
 - (2) Recessive X-linked
 - (3) Dominant Y-linked
 - (4) Recessive Y-linked
180. In a population that is in Hardy weinberg equilibrium, the frequency of a recessive allele for a certain hereditary trait is 0.30. What percentage of the individual in the next generation would be expected to show the dominant trait ?
- (1) 16% (2) 32% (3) 64% (4) 91%
181. Which pedigree is not possible for sickle cell anaemia?



182. Represented below is the inheritance pattern of a certain type of traits in human. Which one of the following condition could be an example of this pattern ?



- (1) Haemophilia
 - (2) Thalassemia
 - (3) Colourblindness
 - (4) Hypertrichosis
183. In Snapdragon, pure red flowered plant are crossed with pure white flowered plant and F_1 individual have pink flowers. Which one of the following is correct for F_1 organisms ?
- (1) Pink flowered plant will produce only pink flowered offsprings
 - (2) They will produce two types of phenotypes & three types of genotypes when selfed.
 - (3) Only one fourth organisms will be homozygous if they are selfed
 - (4) The alleles of hybrid pink flowered plant will segregate, if self pollinated
184. Read the following statements :-
- (a) It produces disorder more often in female than in males.
 - (b) All female off springs will exhibit disorder, if father possesses the same.
 - (c) Do not transmitted to son, if mother does not exhibit disorder.
- Which of the following gene will have the above stated features ?
- (1) Autosomal dominant gene
 - (2) Autosomal recessive gene
 - (3) X-linked dominant gene
 - (4) X-linked recessive gene

185. A normal woman (whose father is colour blind but mother is homozygous (normal) marries a haemophilic man with hypertrichosis. What percentage of progeny will show genotypically any two of the traits out of the three mentioned above at a given time ?

(1) 50% (2) 25% (3) 0% (4) 75%

SECTION-B (BIOLOGY-II)

186. Fill in the blanks in the different columns of the table given below ?

Disease	Casual organism and part it inputs	Medium of Transfer
(i) Common cold	A	Droplets from sneezing of infected person
(ii) B	<i>Trichophyton</i> , Skin, Nails and Scalp	Using towels of infected individuals
(iii) <i>Pneumonia</i>	C	Droplets/ aerosols released by infected person
(iv) <i>Filariasis</i>	<i>Wuchereria</i> , Lymphatic vessels	D

- (1) A. Rhinovirus, Nose and respiratory passage but not lungs
B. Ringworm
C. *Haemophilus influenzae*, Alveoli of lungs
D. Contaminated food and water
- (2) A. Coryza virus, Alveoli of lungs
B. Ringworm
C. *Streptococcus pneumoniae*, Nose and respiratory passage
D. Culex mosquito
- (3) A. Rhinovirus, Nose and respiratory passage but not lungs
B. Ascariasis
C. *Haemophilus influenzae*, Lungs
D. Culex mosquito
- (4) A. Rhinovirus, Nose and respiratory passage but not lungs
B. Ringworm
C. *Haemophilus influenzae*, Alveoli of lungs
D. Female Culex mosquito

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187. Which of the following statement is not correct about typhoid ?

(a) Sustained high fever (39° to 40° F)
(b) Weakness
(c) Constipation
(d) Headache and loss of appetite
(e) Coughing

(1) a, c, d (2) c, d, e
(3) a and e only (4) All are correct

188. How many of the following are responsible for transmission of HIV infection ?

(a) sexual contact with infected person
(b) by sharing infected needles
(c) from infected mother to her child through placenta
(d) Shaking hands
(e) Touch
(f) Through body fluids

Options :-

(1) Two (2) Five (3) Four (4) Six

189. Find out the "incorrect" statement about Tobacco addiction and its effects :-

(1) It's prolonged abuse can lead to cancer of lungs and urinary bladder.
(2) It can be chewed, smoked and snuffed.
(3) It increases carbon mono oxide content in the blood as well as decrease the concentration of haem bound oxygen.
(4) Tobacco does not contain Nicotine

190. **Assertion** : Subsequent encounter with the same pathogen elicits a highly intensified secondary or Anamnestic response.

Reason : Secondary response based on memory of 1st encounter of pathogen.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

191. **Statement A** : Metastasis is the most feared property of benign tumor.

Statement B : Normal cells show a property called contact inhibition.

- (1) Statement A is correct, Statement B is incorrect
- (2) Statement A is incorrect, Statement B is correct
- (3) Both are correct
- (4) Both are incorrect

192. Pneumonia is not caused by

- (1) Streptococcus pneumoneae
- (2) Salmonella typhi
- (3) Haemophilus influenzae
- (4) 1 and 3 both

193. When we make the selfing of heterozygous round (Bb) seed shape Pea plant, then in their progeny in what fraction we obtain large starch grain size seed :

- (1) $\frac{1}{2}$
- (2) $\frac{3}{4}$
- (3) $\frac{1}{4}$
- (4) $\frac{2}{3}$

194. When more than one phenotypes are influenced by the same gene, then dominance is not an autonomous feature of a gene. It also depends on :-

- (1) Gene product
- (2) Gene position
- (3) The particular phenotype that we choose to examine
- (4) (1) and (3) both

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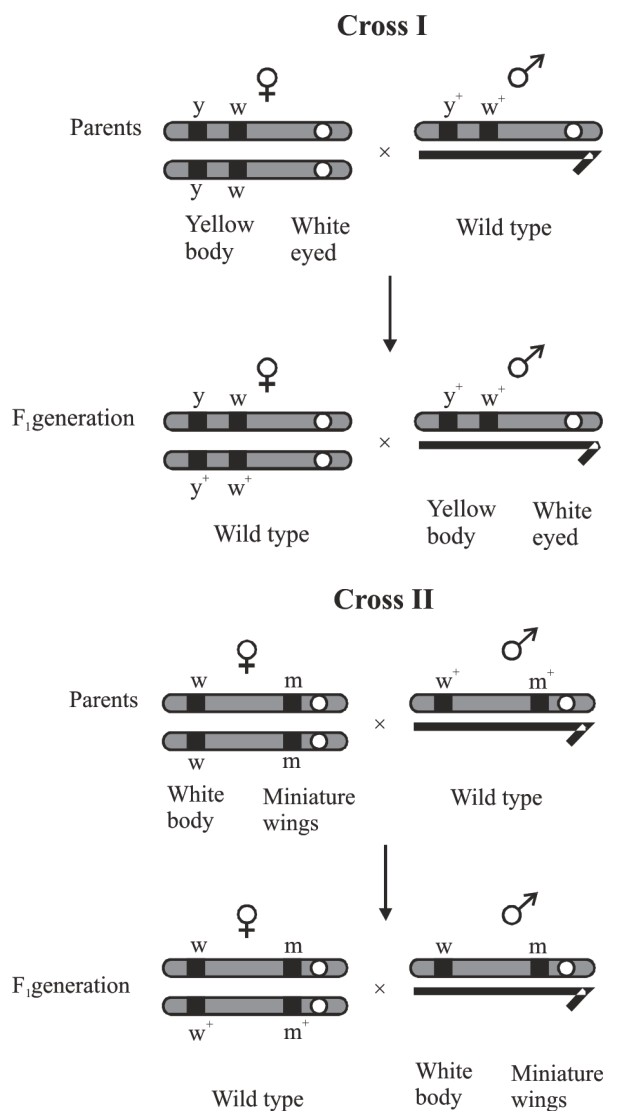
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195. Morgan coined the term to describe the physical association of genes on a chromosome and the term to describe the generation of non parental gene combinations :-

- (1) Recombination, Linkage
- (2) Recombination, Non-recombination
- (3) Linkage, Non-recombination
- (4) Linkage, Recombination

196. Crosses shown in the figure were carried out by Morgan on Drosophila. What percentage of parental offsprings were obtained in F₂ generation in cross-I and cross-II respectively?



- (1) 62.8% and 98.7%
- (2) 1.3 % and 37.2%
- (3) 98.7% and 62.8%
- (4) 37.2% and 1.3%

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197. F_1 wild female was crossed with recessive male and progeny was produced as follows :

$++/yw$	= 202
yw/yw	= 198
$+w/yw$	= 798
$y+/yw$	= 802

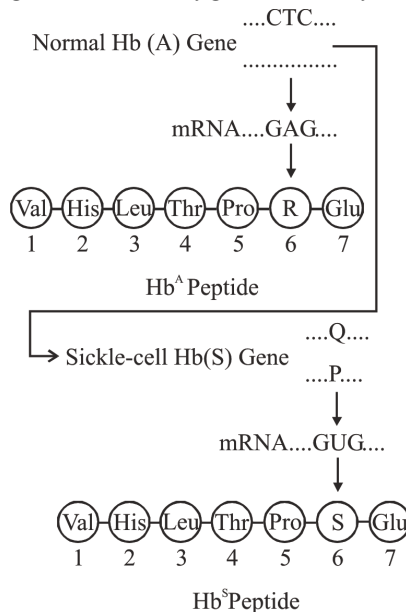
Now match correct combination :-

	Type of inheritance	Strength of linkage	Distance between y and w	Type of arrangement
(1)	Maximum crossing over	80%	20 cM	Cis
(2)	Incomplete linkage	90%	10 cM	Cis
(3)	Incomplete linkage	80%	20 cM	Cis
(4)	Incomplete linkage	80%	20 cM	Trans

198. In a family both parents are carrier for albinism and both have blood group AB. What will be probability of Albinic and type A blood group children :-

- (1) $\frac{3}{16}$
 (2) $\frac{6}{16}$
 (3) $\frac{1}{16}$
 (4) $\frac{4}{16}$

199. Sickle cell anaemia is an autosomal recessive trait that can be transmitted from parents to the offspring when both partners are carrier for the gene or heterozygous. Identify, P, Q, R and S :-



	P	Q	R	S
(1)	GTG	CAC	Valine	Glutamic acid
(2)	GGG	CCC	Glutamic acid	Valine
(3)	GCG	CGC	Valine	Glutamic acid
(4)	GTG	CAC	Glutamic acid	Valine

200. Given below are some organisms

- I – Human male
 II – Female hen
 III – Male Drosophila
 IV – Male grasshopper
 V – Male birds

Suppose :

Male heterogamety = A

Female heterogamety = B

Male homogamety = C

On the basis of sex determination, which of the following combinations is correct ?

- | | A | B | C |
|-----|---------|----|-----|
| (1) | I, IV | II | V |
| (2) | II, IV | I | III |
| (3) | III, IV | II | I |
| (4) | II, V | I | II |

 QUESTIONS BREAK-UP CHART FOR MINOR TEST FROM CURRENT & PREVIOUS UNITS (Session : 2023 - 2024)			
ENTHUSE ADV. COURSE : MEA & MEPS			
MINOR TEST NO. & DATE	CURRENT UNIT	PREVIOUS UNIT (COMPLETE TOPIC)	
07 13-08-2023	SYLLABUS COVERED FROM 17-07-2023 TO 12-08-2023	P	SHM, WAVE MOTION
		C	SOLUTION
		B	ORIGIN, EVOLUTION, DOMESTICATION OF ANIMALS

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